



Lecture # 01

Technical Report Writing

Course Code: **HS-201**

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“Technical Report Writing Today 10th Edition”

The Technical Report

- Intent of a technical report is to communicate an idea/problem to a reader effectively
- A **technical report** (also **scientific report**) is a document that describes the process, progress, or results of technical or [scientific research](#) or the state of a technical or scientific research problem.
- The “Essay” of the scientific world
- State the idea/problem
- Frame your response
- Respond with support for your argument
- Conclude

Why is a Good Report Important?

- **Need to communicate ideas to an audience**
- Knowledge and skills are useless if you cannot communicate your ideas
- Collect information, organize it, and present it in a logical and concise form
- Report must convey the exact meaning you intend
- **Well written reports will help your career and Poorly written reports undermine your credibility and frustrate your reader (i.e., ME!!!)**

How to write a good technical report

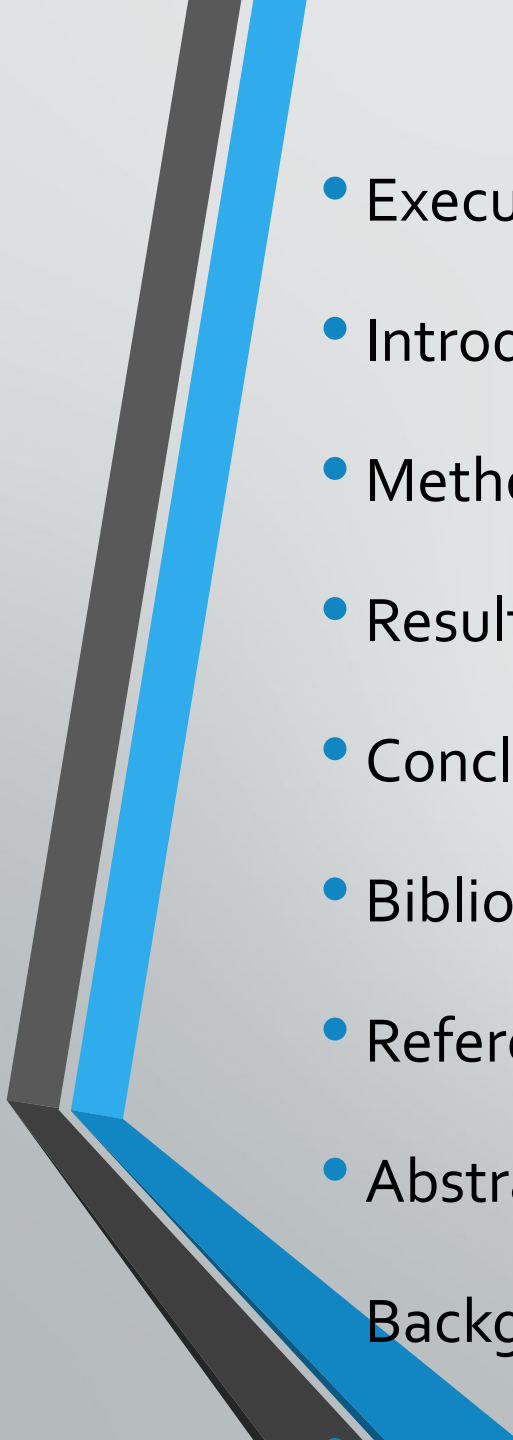
- Before writing the first word:
- Make your mind regarding the message you want to convey
- Try to define the likely audience:
 - Technical audience
 - Non-technical, e.g., general public
- Taking into account the audience's limitations and the message you want to convey, choose an appropriate outline

Report Presentation

- **Binding**
- Permanency (staple, duotang, cerulux, 3-ring)
- Do not submit “loose-leaf” or in folders
- Allow for binding on left margin (“gutter”)
- Keep about 1” (2.5 cm) of white space around page edges

Report Presentation

- **Headings**
- K.I.S.S. principle (avoid boxes and wild fonts)
- Want report to be easy to follow for reader



Technical Report Content

- Executive Summary
- Introduction
- Methodology
- Results/Analysis of
- Conclusions
- Bibliography
- References
- Abstract
- Background

BROAD DIVISION	INDIVIDUAL SECTIONS
1. Preliminary Material	Title of Report
	Table of Contents
	Abstract / Synopsis
2. Body of Report	Introduction
	Literature Review
	Methodology
	Results
	Discussion
	Conclusion
	Recommendations
3. Supplementary Material	References / Bibliography
	Appendices

Executive Summary

- A summary, tailored to high level executives outlining the major findings of the report, i.e., the bottom line. Standalone, not part of main document.

Abstract

- A summary of major research/methodological contributions. Used in research papers and documents.



Background

- A description of the history behind that particular problem. It may cover previous works on the area and previous attempts to solve the problem.

Introduction

- A transition toward the main body of the document. It should take an uninformed reader from a level of zero-knowledge to a level in which the reader is able to understand the main body of the document.

Introduction - components

- A good introduction must have:
 - Motivation (i.e., why is it important?)
 - General
 - Specific
 - Background (i.e., what is the history of this issue?)
 - Objectives (i.e., what are you trying to accomplish?)
 - Scope (i.e., what is the focus of your analysis?)
 - Limitations (i.e., what constraints did you face?)
 - Content (i.e., what is in the report?)



Methodology

- A description of the methodological framework you have used in the project, or investigation. It focuses on the theoretical side of the methods.

Analysis of Results

- A description of the results obtained and analysis of the implications associated with main results. It must be supported by figures and tables to facilitate, not to confuse, the reader

Conclusions

- A summary of the major findings you have arrived to in the previous sections. “Conclusions” is not an analysis section.



Recommendations

- Insights into the next steps you recommend to be taken. This must be supported by the analysis and conclusions section of the report.



Bibliography

- A listing of books and articles you have used, or consulted, for methodological issues.

References

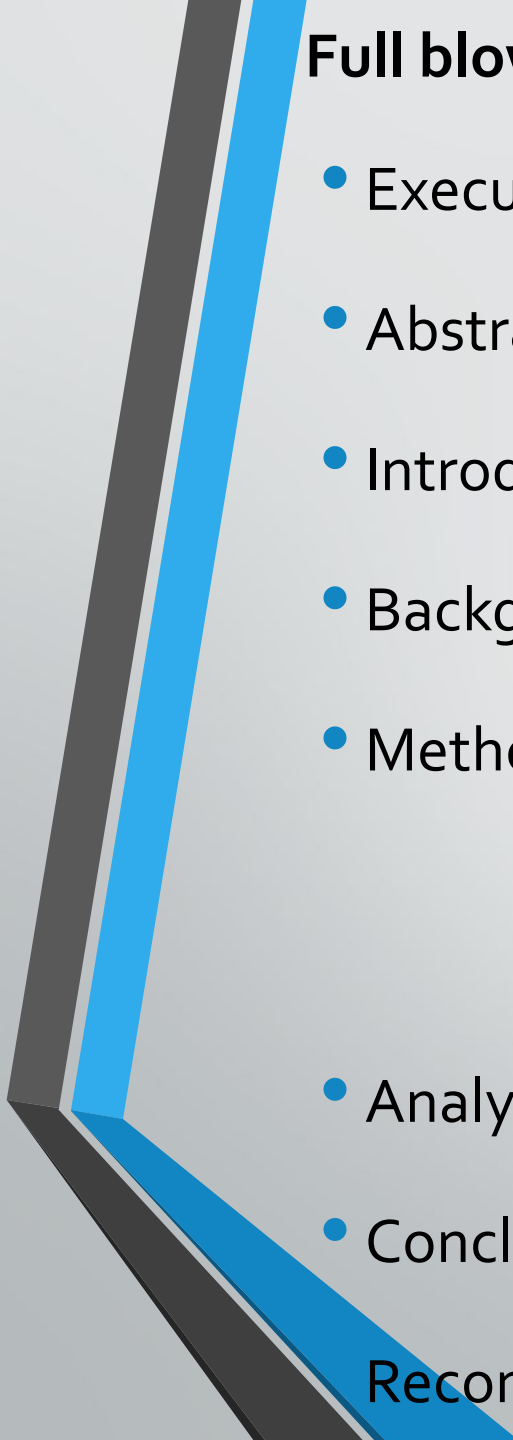
- A listing of books and articles you have used, or consulted, for methodological and non methodological issues.
- Must follow the Harvard Referencing Style
<http://library.curtin.edu.au/referencing/harvard.pdf>

Figures and Tables

- They are intended to facilitate understanding of the document by presenting relevant information and data in a easy-to-understand-way. They must be integrated to the main body.

Appendices

- Intended to present data and analysis, that though important, may not be directly relevant to the main body. It is intended for interested readers only. Appendices are not dump places and they must be classified and organized.
- In books and papers, an appendix is an add-on at the very end of additional information that could be useful to the reader but is not an actual part of the body of the work.



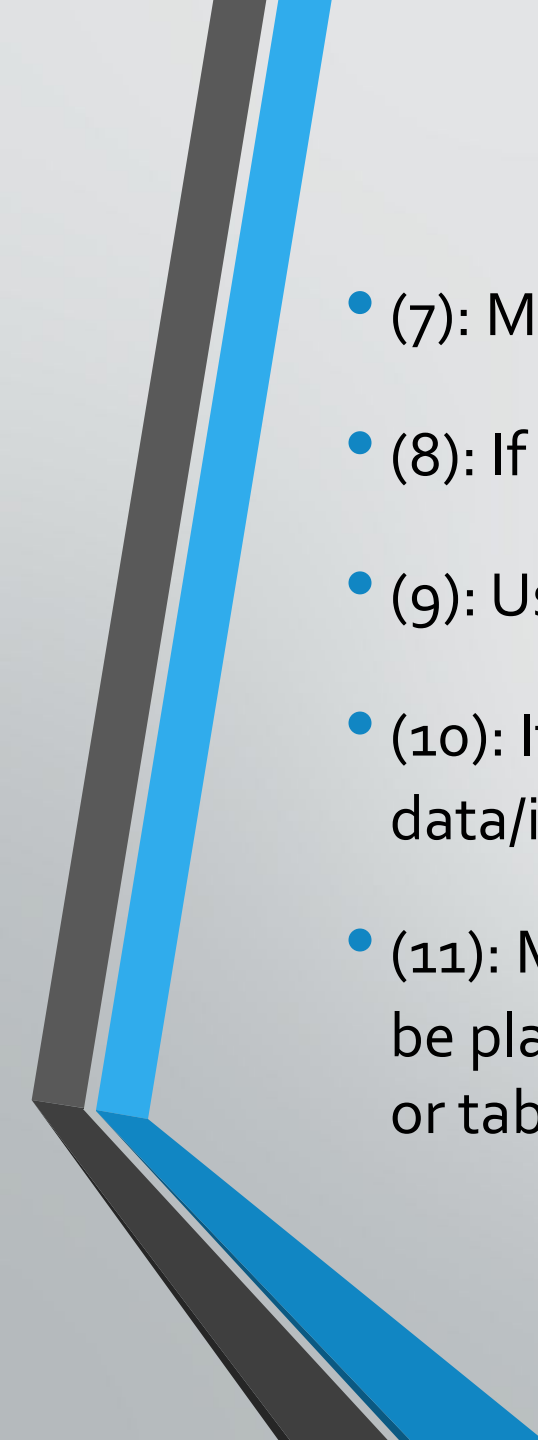
Full blown report:

- Executive Summary (1)
- Abstract (2)
- Introduction (3)
- Background (4)
- Methodology (5)
- Analysis of Results (6)
- Conclusions (7)
- Recommendations (8)

Typical outline

Notes

- (1) and (2): not always needed.
- (3): although not always a section entitled “Introduction” is needed (e.g., in short reports), an introductory section (e.g., a couple of paragraphs) is always required.
- (4): required when the history of the problem (or methodology) is long. Otherwise, include as part of the introduction.
- (5) and (6): Must be separate sections when they are relatively long. Otherwise describe (5) before (6) in the same section.

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- (7): Must follow from the main body (must be supported by).
 - (8): If short, put at the end of conclusions.
 - (9): Use one or the other.
 - (10): It must be classified and organized to present important data/information not directly relevant to the main body of document.
 - (11): Must be integrated to main body of document, figures and tables must be placed after they are first referenced. If no reference is made to a figure or table, why did you put it in the main body?



End